

Which social determinants best predict maternal morbidity in the United States?

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Abstract

Maternal morbidity remains a significant public health concern in the United States despite the country's substantial healthcare spending and resources. This study examines which social determinants best predict maternal morbidity and whether their relative importance changes across analytical scales. Using CDC WONDER natality data from 2023 through mid-2026, this analysis applies OLS regression and machine-learning approaches, including gradient boosting and CatBoost, to examine state-level and more granular demographic data. Results indicate that no single social determinant consistently emerged as the strongest predictor. Teen birth rate was the only statistically significant predictor in the state-level OLS model, while educational disadvantage and self-pay status were most important in gradient boosting. At the more granular level, race and Hispanic origin emerged as prominent predictors in separate analyses. These findings suggest that maternal morbidity reflects interconnected social conditions and that effective public health interventions may need to be tailored to specific populations and contexts.

Keywords: maternal morbidity; maternal mortality; maternal health; birth outcomes

1. Introduction

Maternal morbidity is defined as experiencing a short- or long-term physical or mental health problem, illness, or disability that results from or is aggravated by pregnancy, childbirth, or the postpartum period. Pregnancy is a transformative and life-altering experience that is already difficult, and social factors that, for the most part, are outside of a mother's and child's control should not contribute to adverse birth and health outcomes. The maternal morbidity factors that I measure for in my dataset are ICU admission, transfusion, perineal laceration, unplanned hysterectomy, and uterine rupture. Overall, maternal morbidity is associated with a decrease in quality of life, can cause long-term damage, and carries a heavy financial and emotional burden for affected mothers.

Despite the United States being the wealthiest country in the world and spending more on healthcare per capita than other high-income countries, it has exceptionally poor maternal health outcomes compared with its wealthy peers. According to the Commonwealth Fund, the United States has an overall maternal mortality rate of 18.6 per 100,000 live births, and most notably, Black women have a mortality rate of 50.3 per 100,000 live births [?]. This apparent contradiction between national wealth and maternal health outcomes raises questions about whether the healthcare system alone is sufficient to protect mothers from adverse outcomes.

Social determinants of health is a coined public health term that describes non-medical factors that affect the quality of life of individuals, such as race, ethnicity, education, age,

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insurance, and urban/rural residence. Despite social determinants of health not being direct medical conditions, these factors may affect people's access to prenatal care and healthcare resources, their ability to respond to complications, the quality and accessibility of healthcare, socioeconomic resources, and exposure to chronic stress. Social determinants of health can have a large impact on quality of life and health outcomes, potentially leading to unequal health and birth outcomes for mothers and their children.

It is important to note that many social factors are related to one another; for instance, education and teen birth rates, or income and healthcare insurance. Because these factors can overlap, looking at each social determinant individually may not show which factors provide unique information about maternal morbidity. Therefore, it is important to establish which factors provide the greatest predictive information about maternal morbidity when multiple social determinants are considered simultaneously.

The goal of this project is to find which social determinants best predict maternal morbidity in the United States, as well as to determine whether the relative importance of these determinants' changes depending on the level of analysis. To investigate this question, this study uses CDC WONDER natality data from 2023 through mid-2026 and compares conventional regression with machine-learning approaches. The analysis first examines state-level variation across 51 states and Washington, D.C., before returning to more individualized demographic combinations contained in the underlying data.

2. Literature

2.1. Weathering Hypothesis

First, the weathering hypothesis [1] was developed specifically to explain the persistent, well-documented racial gap in Black maternal health outcomes in the United States. The hypothesis argues that chronic exposure to social, economic, and psychosocial stressors accelerates biological aging and physiological decline among Black women, producing health disparities that show up in pregnancy and childbirth, specifically because pregnancy places acute stress on an already-taxed body. Critically, weathering theory does not predict that racial disparities will disappear once income or education are accounted for. A well-established and often-cited finding in this literature is that highly educated, higher-income Black women still show elevated rates of severe maternal morbidity and mortality relative to their White counterparts, suggesting that the mechanism is not simply socioeconomic. Given this framework, race was expected to emerge as one of the strongest predictors in this analysis, independent of other factors. This theory therefore provides a specific prediction for the present analysis: if weathering processes contribute to maternal morbidity, race should remain an important predictor even after accounting for other socioeconomic and demographic factors. This expectation is particularly relevant to the present study because the analysis considers race alongside education, insurance, age, and other social determinants that may capture different dimensions of maternal health disparities.

2.2. Fundamental Cause Theory

The second theoretical framework is Fundamental Cause Theory [2], which argues that socioeconomic status functions as a "fundamental cause" of health disparities because it provides flexible resources, including money, knowledge, power, prestige, and beneficial social connections, that can be redeployed to avoid health risks regardless of which specific disease or mechanism is currently most threatening. Applied to maternal health, this theory predicts that income and insurance coverage should matter a great deal, since financial resources can influence access to prenatal care, choice of delivery facility, and the ability to act quickly on early warning signs of complications. This motivated the expectation that self-pay or uninsured status and Medicaid coverage would show strong

associations with morbidity. More broadly, Fundamental Cause Theory suggests that socioeconomic indicators should retain predictive importance even when the specific mechanisms contributing to maternal morbidity vary across populations. Thus, education and insurance-related measures were expected to be among the stronger predictors in the analysis.

2.3. *Connecting Frameworks*

Together, these theories suggest that maternal morbidity may be influenced by both racialized experiences and access to socioeconomic resources. However, these factors are not independent of one another. Race, education, insurance coverage, age, and other social determinants can be closely related, making it difficult to determine which factors provide unique information about maternal morbidity when considered separately. This overlap motivates the use of multiple modeling approaches to examine the relative predictive importance of these factors simultaneously. Rather than if either race or socioeconomic status alone explains disparities in maternal morbidity, this analysis tests whether the patterns predicted by both theoretical frameworks are reflected in the data. 2.4 Research Gap While both frameworks provide explanations for disparities in maternal health, they emphasize different mechanisms through which social conditions may influence outcomes. The weathering hypothesis emphasizes the physiological consequences of racialized social stress, while the Fundamental Cause Theory emphasizes the role of socioeconomic resources. Examining both frameworks within the same analysis allows this study to assess whether race, socioeconomic characteristics, and other social determinants provide predictive information about maternal morbidity. Additionally, comparing conventional regression with machine-learning approaches allows the analysis to examine whether the relative importance of these factors changes depending on how relationships between predictors and maternal morbidity are modeled.

3. **Data**

3.1. *Data Source: CDC WONDER Natality Database*

This analysis draws birth-level data from the CDC WONDER Natality Database, covering births from 2023 through mid-2026. The Natality Database contains extensive information on maternal, paternal, and infant demographic characteristics recorded on birth certificates. These data provide information on maternal characteristics, including race, age, education, ethnicity, insurance coverage, and urbanicity, as well as information on maternal morbidity.

3.2. *Query 1*

Due to CDC WONDER’s five-variable limit per query, I extracted the data using two separate queries. Query 1 was grouped by State × Race of Mother (6) × Age of Mother (10) × Mother’s Education × Maternal Morbidity Checked. This query allowed me to examine how maternal morbidity varied across states and combinations of maternal race, age, and education.

3.3. *Query 2*

For the second query, I grouped the data by State × Urbanicity (Metro/Nonmetro) × Hispanic Origin × Source of Payment for Delivery × Maternal Morbidity Checked. This query allowed me to incorporate additional measures of geographic context, ethnicity, and insurance/payment status that could not be included in the first query due to CDC WONDER’s variable limit. 3.4 Dependent Variable Both queries were cross tabulated against a "Maternal Morbidity Checked" indicator. In the CDC WONDER database, this

indicates that the birth certificate recorded at least one of five severe maternal morbidity complications: ICU admission, transfusion, perineal laceration, unplanned hysterectomy, or uterine rupture. Nevertheless, it is important to note that maternal morbidity in this analysis is a documentation-based measure rather than a direct clinical measurement of maternal health. The indicator reflects whether a complication was recorded on the birth certificate and therefore may be influenced by differences in documentation or reporting practices.

3.4. State-Level Data

The two WONDER queries were initially collapsed to the state level and merged into a single dataset containing 51 observations, representing the 50 states and Washington, D.C. From these data, I calculated 11 composite demographic and socioeconomic predictors representing race, ethnicity, age, education, insurance/payment status, and geographic context. Variable Concept

3.5. Granular Demographic Data

Because the state-level dataset contained only 51 observations but 11 predictors, the analysis later evolved to a more granular demographic combination contained in the original WONDER queries. This allowed the analysis to move beyond state averages and examine whether patterns in maternal morbidity changed when demographic characteristics were considered simultaneously within states. The first query ($n = 14,561$) produced substantially more demographic combinations than the second query ($n = 1,939$) providing additional observations for the later classification analyses.

4. Methodology

4.1. Exploratory Analysis

To begin, I examined the distributions of the predictors, correlations between variables, relationships among predictors, and variation across states. The analysis found that several predictors were significantly associated with maternal morbidity when examined individually. In the initial analysis, teen birth rate, bachelor's degree attainment, private insurance, less than high school education, and Medicaid showed significant relationships with maternal morbidity. However, when the predictors were considered simultaneously in the multivariate OLS model, only teen birth rate remained statistically significant. This suggests that several predictors may contain overlapping socioeconomic information.

4.2. OLS Regression

An OLS regression was used to estimate the association between state-level maternal morbidity and multiple social determinants while holding the other predictors constant. Each standardized OLS coefficient represents the estimated change in state-level maternal morbidity associated with a one-standard-deviation increase in the predictor, while holding the other predictors constant. This allowed me to evaluate whether individual social determinants were associated with maternal morbidity after accounting for the other factors included in the model.

4.3. Gradient Boosting / CatBoost

Because OLS assumes a linear relationship between the predictors and outcome, it may not capture nonlinear relationships or more complex interactions between social determinants. To account for these possibilities, I incorporated gradient-boosting models into the analysis. For the initial state-level analysis, I used a Gradient Boosting Regressor to predict the state-level maternal morbidity rate. Feature importance was evaluated using

permutation importance, which measures how much model performance decreases when the values of a particular predictor are randomly shuffled. A larger decrease indicates that the model relied more heavily on that predictor when making its predictions. For the more granular demographic-combination analysis, I used CatBoost classification models. Feature importance was evaluated using SHAP values, which estimate how much each variable contributes to the model's predictions. The mean absolute SHAP value was used to rank the overall importance of each predictor, with larger values indicating greater influence on the model's predictions. The CatBoost models were used alongside logistic regression to determine whether the relative importance of predictors was consistent across modeling approaches.

4.4. Model Evaluation

The initial state-level OLS model produced an in-sample R^2 of 0.373, meaning that the model explained approximately 37.3 % of the variation in state-level maternal morbidity. The Gradient Boosting model produced a higher in-sample R^2 of 0.752. However, the higher in-sample performance did not translate to better out-of-sample prediction. After cross-validation, both models produced negative R^2 values, meaning that their predictions performed worse than simply predicting the average maternal morbidity rate. Therefore, neither state-level model generalized well to new observations. Because of these limitations, I returned to the more granular demographic-combination data for the final stage of analysis.

4.5. Granular Demographic-Combination Analysis

Because the state-level dataset contained only 51 observations, a second analysis returned to the more specific demographic combinations contained within the original CDC WONDER queries. Query 1 crossed State, Race, Age, Education, and Maternal Morbidity, resulting in 14,561 demographic combinations. Query 2 crossed State, Urbanicity, Hispanic Origin, Insurance, and Maternal Morbidity, resulting in 1,939 demographic combinations. The larger number of observations allowed me to examine whether the patterns identified in the state-level analysis remained consistent at a more granular level. For each query, I used logistic regression as a conventional statistical model and CatBoost as a machine-learning approach. Comparing the two models allowed me to determine whether the relative importance of predictors was consistent across modeling approaches. CatBoost achieved an AUC of approximately 0.82 for Query 1 and 0.65 for Query 2, indicating that both models performed better than random classification, although Query 1 demonstrated stronger predictive performance.

5. Results

This section presents results in the order of the analytical hierarchy described in the methodology: first the state-level ecological models (Sections 5.1–5.4), then the individual-record models estimated on the two Vital Statistics queries (Sections 5.5–5.6), and finally the state-stratified models that test whether these relationships hold uniformly across geographic contexts (Section 5.7).

5.1. Predictors Are Highly Interconnected

Before modeling, I examined the distributions of the predictors and the relationships between social determinants. Several predictors showed relationships with maternal morbidity when examined individually, but many of the social determinants were also correlated with one another. For example, teen birth rate and bachelor's-degree attainment showed a strong inverse relationship, meaning that states with higher rates of teen births tended to have lower rates of bachelor's-degree attainment.

This is important because correlated predictors may capture overlapping dimensions of socioeconomic conditions rather than completely independent effects. Therefore, a predictor that appears strongly related to maternal morbidity on its own may become less important when other related social determinants are considered simultaneously.

Table 1. Correlation Between Social Determinants and Maternal Morbidity

Predictor	Correlation	p-value
Teen birth rate	-0.4842	0.0003
Bachelor’s degree attainment	0.4296	0.0017
Private insurance	0.3954	0.0041
Less than high school	-0.3808	0.0058
Medicaid	-0.2911	0.0382
Advanced age	0.2719	0.0536
Self-pay	-0.2623	0.0629
Hispanic	-0.2570	0.0687
White	0.1070	0.4550
Black	-0.0727	0.6123
Nonmetro	0.0186	0.8969

5.2. Teen Birth Rate Was the Only Significant OLS Predictor

I first used an ordinary least squares (OLS) regression to examine the relationship between state-level maternal morbidity and the 11 social determinants while holding the other predictors constant (Figure 1). In this model, only percentage of teen births was statistically significant (coefficient = −0.0053, p = 0.03), while all other predictors had p-values greater than 0.05.

The negative coefficient indicates that, after accounting for the other predictors in the model, states with higher teen birth rates tended to have lower overall documented maternal morbidity rates. In practical terms, a negative coefficient means that the predicted morbidity rate decreased as the teen birth rate increased, while the other variables were held constant.

This finding is counterintuitive and differs from the individual-level age patterns observed later in the analysis. Importantly, this result describes a state-level association and does not mean that individual teenagers are less likely to experience maternal morbidity.

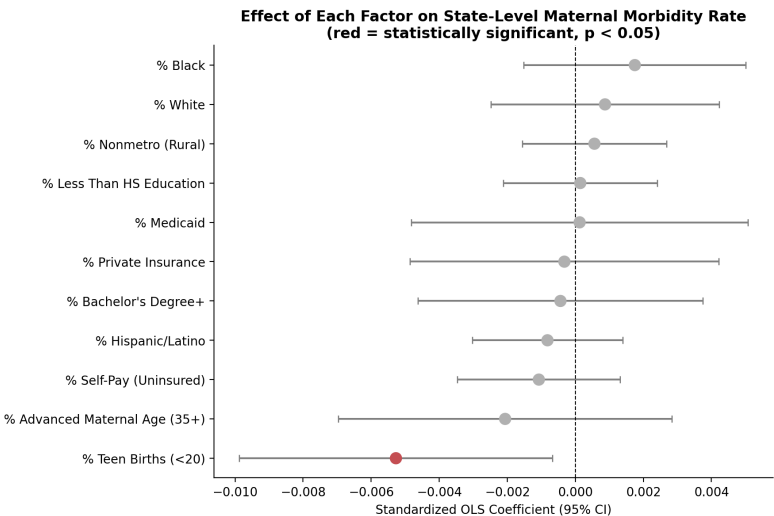


Figure 1. Standardized OLS coefficients with 95% confidence intervals for the state-level model of severe maternal morbidity. The red marker indicates the only statistically significant predictor ($p < 0.05$); the dashed vertical line marks zero.

5.3. Machine Learning Produces a Different Ranking

Next, I used gradient boosting to determine whether a model capable of capturing more complex and potentially nonlinear relationships would identify the same predictors as the OLS model. Unlike OLS, which estimates linear associations and tests whether individual coefficients are statistically significant, gradient boosting evaluates how useful different predictors are for making predictions.

The gradient boosting model identified *less than high school education* as the most important predictor, followed by self-pay status and teen birth rate. This differed from the OLS results, in which teen birth rate was the only statistically significant predictor.

Table 2 and Figure 2 demonstrate an important distinction between statistical significance and predictive importance. Permutation importance measures how much model performance changes when the values of a predictor are randomly shuffled. A larger decrease in model performance indicates that the predictor was more important to the model's predictions. In this analysis, less than high school education had the largest permutation importance (0.2111), followed by self-pay status (0.1264) and teen birth rate (0.0961).

These results demonstrate that a predictor can contribute substantially to a model's predictive performance without being statistically significant in an OLS model. Conversely, a predictor that is statistically significant in a linear model does not necessarily have the greatest predictive importance in a machine-learning model.

Table 2. Gradient Boosting Permutation Feature Importance

Predictor	Importance	Standard Deviation
Less than high school	0.21111	0.05439
Self-pay	0.12644	0.03522
Teen birth rate	0.09607	0.02432
Hispanic	0.09243	0.01806
Bachelor's degree attainment	0.07722	0.02474
Private insurance	0.06364	0.01813
Nonmetro	0.05858	0.01332
Black	0.05226	0.01618
Advanced age	0.01443	0.00421
White	0.01086	0.00312
Medicaid	0.00291	0.00116

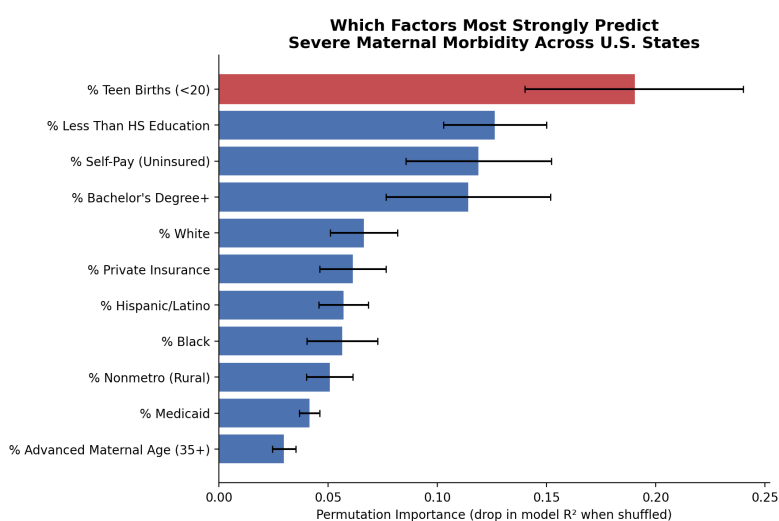


Figure 2. Permutation importance from the gradient boosting model, measured as the reduction in R^2 when each predictor is shuffled. Error bars show variation across permutation repetitions.

5.4. The State-Level Models Overfit

The initial model performance appeared stronger for gradient boosting than for OLS. The OLS model had an in-sample R^2 of 0.373, meaning that the model explained approximately 37.3 % of the variation in state-level maternal morbidity within the data used to fit the model. Gradient boosting had a substantially higher in-sample R^2 of 0.752.

However, in-sample performance does not indicate how well a model will perform on new data. To evaluate generalizability, I used cross-validation, which repeatedly trains the model on one portion of the data and evaluates its predictions on observations that were not used for training.

After cross-validation, both models produced negative R^2 values. The cross-validated R^2 was -0.418 for OLS and -0.054 for gradient boosting. A negative R^2 indicates that the model performed worse on the held-out observations than a simple model that predicted the average morbidity rate for every observation.

Therefore, although gradient boosting appeared substantially stronger than OLS when evaluated on the original data, neither state-level model generalized well to new observations. The combination of only 51 state-level observations and 11 predictors likely limited the amount of information available for the models to identify reliable patterns. The large difference between the in-sample and cross-validated performance also suggests that the models were fitting patterns specific to the available states rather than consistently capturing relationships that could be generalized beyond the dataset.

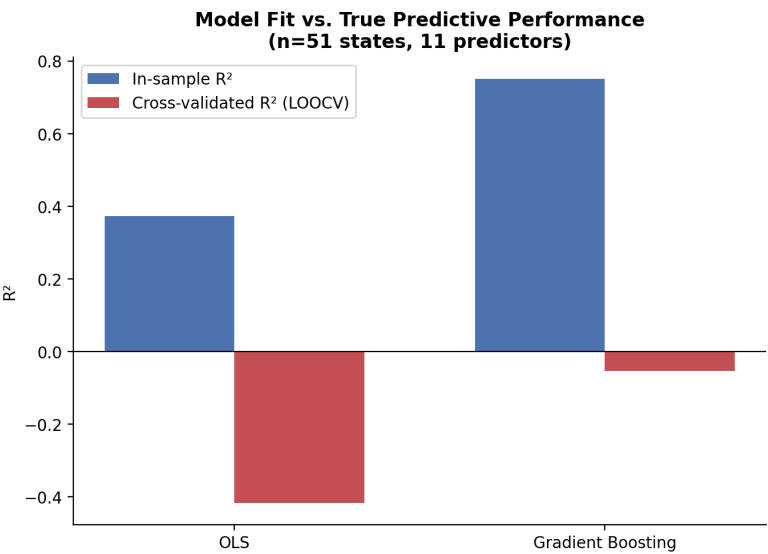


Figure 3. In-sample versus leave-one-out cross-validated R^2 for the OLS and gradient boosting models ($n = 51$ states, 11 predictors). Negative cross-validated values indicate performance worse than predicting the mean.

5.5. Race Becomes the Strongest Predictor at the More Granular Level

Because the state-level analysis contained only 51 observations, I returned to the more granular demographic combinations contained within the original CDC WONDER query. Query 1 contained 14,561 observations and crossed state of residence, mother’s race, mother’s age, education, and maternal morbidity.

The larger number of observations allowed the CatBoost model to identify patterns at a more granular level. As seen in Figure 4, mother’s race was the most influential predictor, followed by state of residence, mother’s education, and age of mother.

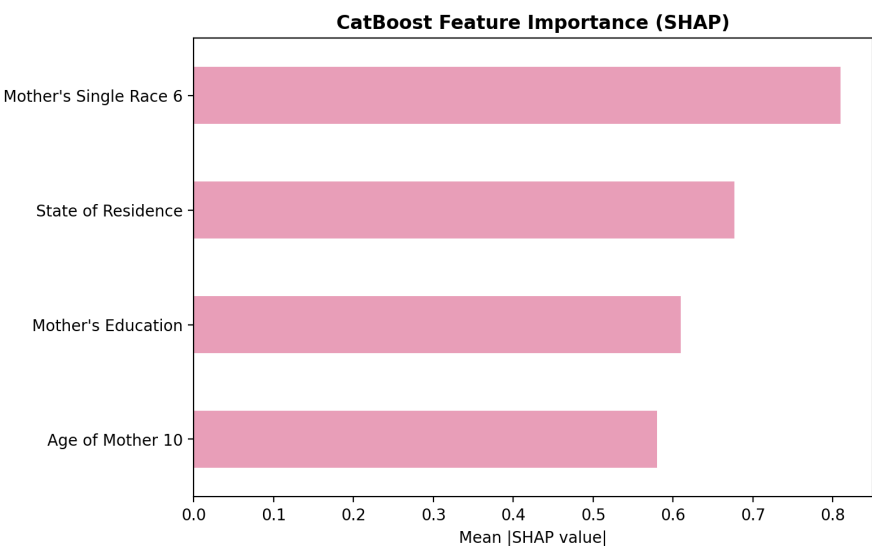


Figure 4. Query 1 CatBoost feature importance, measured as mean absolute SHAP value across 14,561 records.

Figure 5 provides more information about this feature-importance ranking. SHAP values measure how much an individual feature contributed to a specific model prediction. Values farther from zero represent larger contributions, while positive SHAP values indicate that a feature pushed the model toward a higher predicted probability of documented maternal morbidity and negative values indicate that it pushed the prediction toward a lower probability.

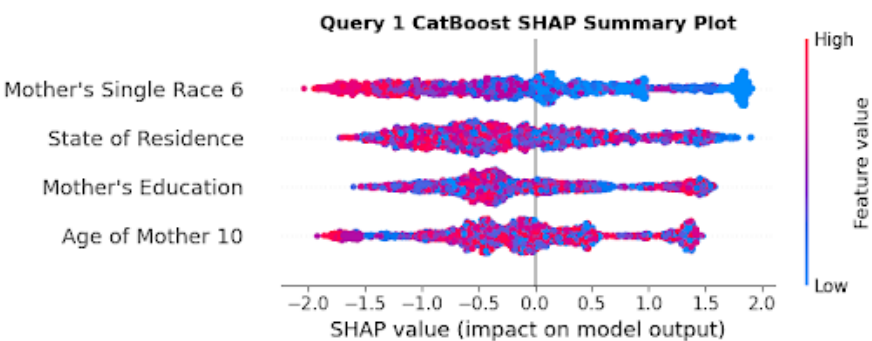


Figure 5. Query 1 SHAP summary plot. Each point is a single record; horizontal position shows that record’s contribution to the predicted log-odds of documented morbidity, and colour indicates the feature value.

Each dot in the beeswarm represents an observation in the test data. The wide horizontal distribution of race indicates that race contributed substantially to variation in the model’s predictions across demographic combinations. The other predictors also show variation in their SHAP values, but their overall distributions are narrower than that of race.

The Query 1 models also demonstrated relatively strong predictive performance. The logistic regression had an AUC of 0.817, while the CatBoost model had an AUC of approximately 0.990. AUC measures a model’s ability to distinguish between observations with and without documented maternal morbidity, with a value of 0.5 representing performance equivalent to random classification and values closer to 1 indicating stronger discrimination.

Figure 6 results additionally indicated that births in the White category had the highest predicted probability of documented morbidity relative to the reference group. This

finding should be interpreted cautiously, as the analysis identifies statistical and predictive relationships rather than demonstrating that race biologically causes maternal morbidity.

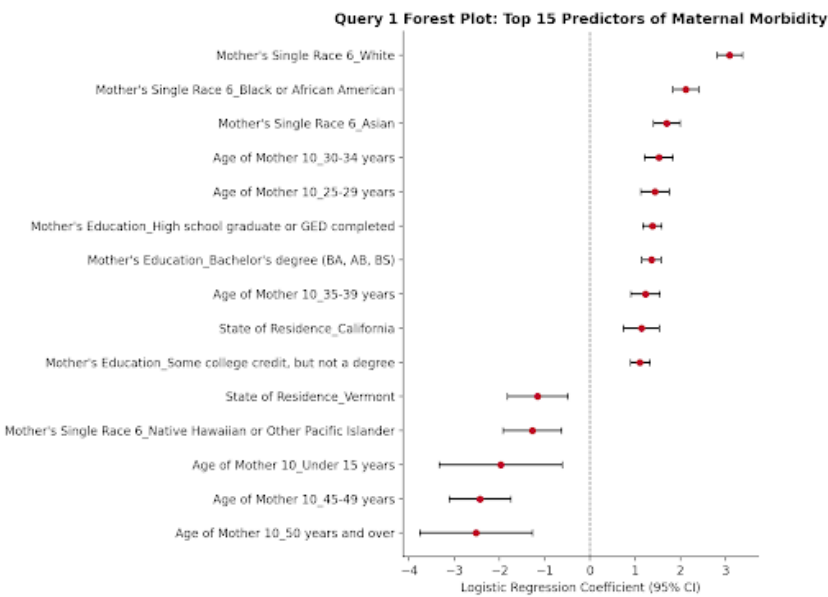


Figure 6. Query 1 logistic regression coefficients with 95% confidence intervals for the fifteen strongest predictors of documented maternal morbidity. Coefficients are relative to the omitted reference category within each variable.

5.6. Query 2 Has a Weaker Predictive Signal

Query 2, which examined urbanicity, Hispanic origin, and source of payment for delivery, produced weaker predictive performance than Query 1. The logistic regression model achieved an AUC of 0.659, compared with 0.817 for Query 1, indicating that Query 2 provided less information for distinguishing between demographic combinations with and without documented maternal morbidity. Nonetheless, its performance was still above the 0.50 level expected from random classification. The CatBoost model achieved a substantially higher AUC of 0.977, indicating that the model was able to capture nonlinear or more complex patterns that were not represented as effectively by logistic regression.

Despite the weaker overall predictive performance, the SHAP analysis (Figure 7) identified mother’s Hispanic origin as the most influential feature, followed by source of payment for delivery, state of residence, and urbanicity.

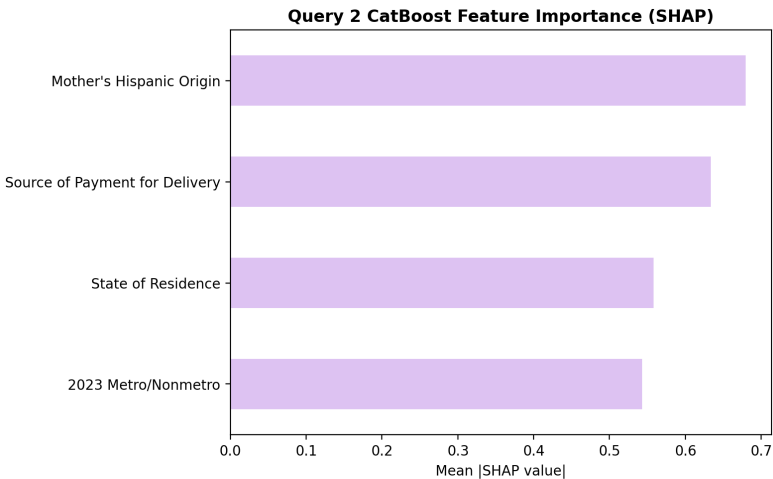


Figure 7. Query 2 CatBoost feature importance (mean absolute SHAP value).

Figure 8 illustrates the distribution of these contributions across individual observations. Mother’s Hispanic origin displays the widest range of SHAP values, indicating that this variable contributed substantially to differences in individual model predictions. Source of payment also showed a relatively wide distribution, while state and urbanicity had comparatively smaller overall distributions.

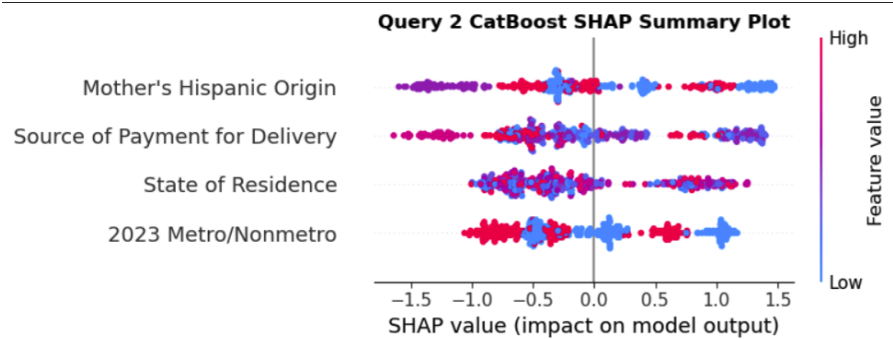


Figure 8. Query 2 SHAP summary plot showing the distribution of individual feature contributions.

Because these predictors are categorical rather than continuous measurements, the color scale in the SHAP plot represents the numerical encoding of categories rather than a meaningful "high" or "low" value. Therefore, the primary interpretation of this plot is the magnitude and direction of each feature’s contribution, rather than interpreting the colors as a continuous increase or decrease in Hispanic origin, insurance status, or state.

The difference between Query 1 and Query 2 is therefore important. Race was the most influential feature among the predictors included in Query 1, while Hispanic origin was the most influential feature among the predictors included in Query 2. However, because the two variables were evaluated in separate models, this does not establish that one is universally a stronger predictor than the other.

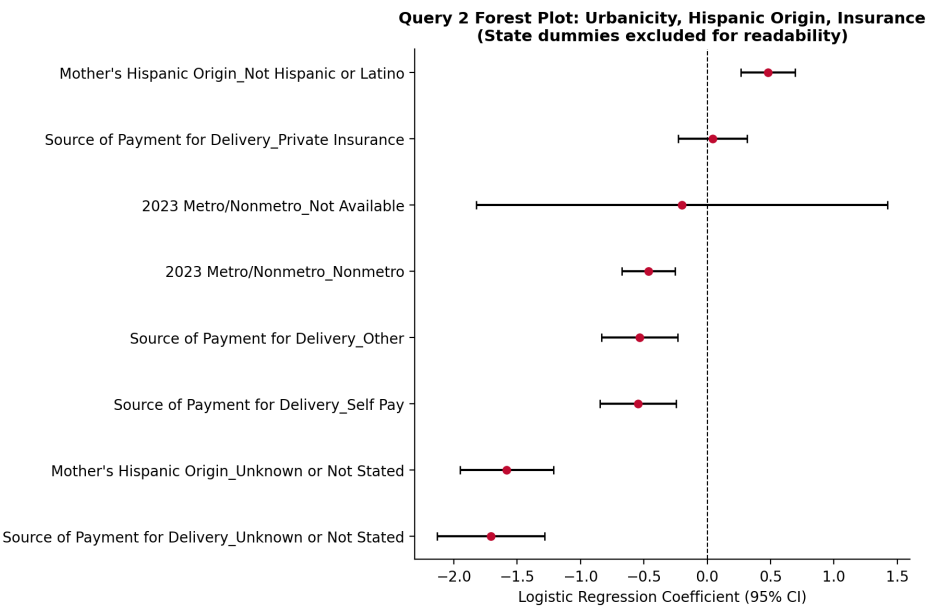


Figure 9. Query 2 logistic regression coefficients with 95% confidence intervals. State dummy variables are estimated but omitted from the plot for readability.

5.7. Effects Vary Across States

To further examine geographic variation, I selected five states representing different regions of the United States: California, New Hampshire, Alabama, Nebraska, and Texas.

The heat map demonstrates that documented maternal morbidity rates differed across these five states, with New Hampshire displaying the highest rate among the selected states.

The purpose of this comparison was not to rank states by maternal health performance, but rather to demonstrate that maternal morbidity is not distributed uniformly across the United States. The state-specific OLS analysis further showed that some social determinants had relatively consistent effects in the same direction across states, while other relationships varied considerably.

The differences seen in Figure 10 and Figure 11 suggest that social determinants may not operate identically across geographic contexts. Rather than indicating that every state has one universal "top" social determinant, the results suggest that the relationship between social conditions and documented maternal morbidity may depend partly on the characteristics and circumstances of individual states.

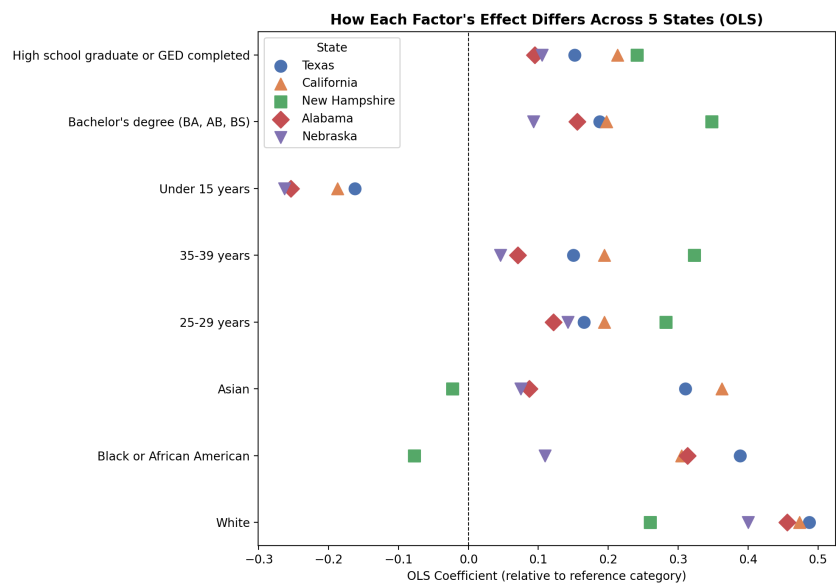


Figure 10. State-specific OLS coefficients for the Query 1 predictors.

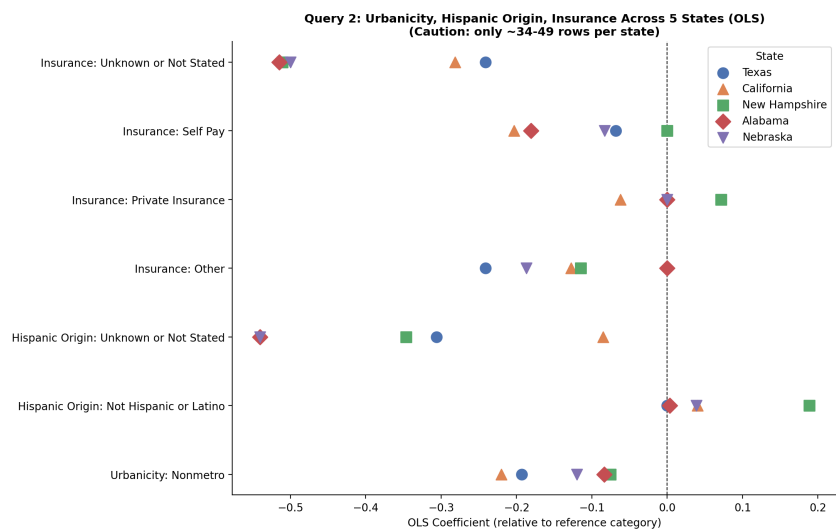


Figure 11. State-specific OLS coefficients for the Query 2 predictors (urbanicity, Hispanic origin, and source of payment). Estimates rest on approximately 34–49 rows per state and should be interpreted with caution.

6. Discussion

6.1. Weathering Hypothesis

According to the Weathering Hypothesis, race should remain an important predictor of maternal morbidity even after accounting for other social and socioeconomic characteristics. In the more granular analysis, race was the strongest factor in Query 1, providing partial support for the weathering framework. However, race was not the dominant predictor at the state level. This suggests that the relationship between race and maternal morbidity may depend on the level at which the data are analyzed.

6.2. Fundamental Cause Theory

Fundamental Cause Theory predicts that socioeconomic resources, including education and healthcare access, should influence health outcomes. The state-level Gradient Boosting model provided some support for this framework, as less than high school education and self-pay status were among the most important features. However, these factors did not all remain statistically significant in the OLS model. Therefore, the findings provide partial rather than definitive support for Fundamental Cause Theory.

6.3. Interpreting Differences Across Analytical Scale

The differences between the state-level and more granular analyses demonstrate that the apparent importance of social determinants depends partly on the level at which maternal morbidity is examined. The state-level finding that teen birth rate was the strongest OLS predictor does not necessarily contradict the prominence of race in the more granular Query 1 analysis. Rather, these findings reflect different patterns of variation within the data. A social determinant that helps distinguish states from one another may not be the same factor that best distinguishes demographic combinations within those states. Consequently, the results should not be reduced to a single "most important" determinant without considering the population and level of analysis being examined.

6.4. Policy Implications

These findings suggest that policymakers should not identify one social determinant and target it universally. Healthcare access, education, insurance coverage, racial disparities, and geographic context may interact differently across populations and states. Maternal-health interventions may therefore need to be tailored to specific populations and contexts rather than organized around one universal "top factor."

7. Limitations

The first limitation, which was particularly relevant to the initial state-level OLS and gradient boosting models, was the small sample size. With only 51 observations and 11 predictors, both models were prone to overfitting. This suggests that these patterns did not generalize well beyond the observations used to train the models.

Moreover, many of the social determinants examined in this analysis are likely correlated with one another. For instance, in my preliminary data analysis, I found that teen birth rate and bachelor's degree attainment were inversely correlated. This overlap makes it difficult to completely isolate the independent effect of any single social determinant, particularly when many social conditions are interconnected. There were also several important social determinants that were not included in this analysis, including poverty rate, median household income, and healthcare workforce shortages. Some variables, such as education and insurance coverage, were used as proxies for broader socioeconomic conditions, but these measures do not fully capture differences in income, access to healthcare,

or available medical resources. As a result, the models may not account for important factors that contribute to maternal morbidity.

Finally, "Morbidity Checked" is a documentation-dependent measure rather than a direct clinical outcome. Patterns that appear to reflect demographic differences in risk may partly reflect differences in how thorough complications are documented across hospitals and states rather than differences in underlying risk alone. Differences in medical practices, documentation, and reporting may therefore affect the accuracy and comparability of the outcome measure. This makes it difficult to determine whether differences identified in the models reflect differences in underlying maternal health risk, differences in healthcare practices, or some combination of the two.

7.1. Future Research

Future research could address these limitations by reducing the number of highly correlated predictors or using statistical approaches specifically designed to account for multicollinearity.

It would also be valuable to examine how adding additional socioeconomic and healthcare-access variables changes the results. Variables such as income, poverty, healthcare workforce availability, and geographic areas with healthcare shortages could provide a more complete representation of the social and structural conditions surrounding maternal health.

Future research should ideally use clinical measures of maternal morbidity rather than documentation-dependent measures. Using more directly measured clinical outcomes could reduce the influence of differences in documentation and medical practices across hospitals and states, providing a more accurate assessment of how social determinants contribute to maternal morbidity.

There were also several important social determinants that were not included in this analysis, including poverty rate, median household income, and healthcare workforce shortages. Some variables, such as education and insurance coverage, were used as proxies for broader socioeconomic conditions, but these measures do not fully capture differences in income, access to healthcare, or available medical resources. As a result, the models may not account for important factors that contribute to maternal morbidity.

8. Conclusion

There is no single universally dominant social determinant of maternal morbidity. Across the different analyses, the relative importance of social determinants changed depending on the level of analysis and modeling approach. At the state level, teen birth rate was the only statistically significant predictor in the OLS model, while gradient boosting highlighted educational disadvantage and self-pay status as important features. However, the poor cross-validated performance of the state-level models demonstrated the limitations of using a small number of state-level observations to predict maternal morbidity.

When returning to the more granular demographic-combination data, race became more prominent in Query 1, while Hispanic origin was more prominent in Query 2. Although Query 2 had weaker predictive performance than Query 1, its AUC of 0.65 still indicates meaningful predictive ability.

Additionally, the differences observed across New Hampshire, Texas, California, Alabama, and Nebraska imply that state-specific context may influence the relationship between social determinants and maternal morbidity.

Overall, these findings suggest that maternal morbidity should be understood as the product of interconnected social conditions rather than one isolated determinant. Determin-

ing which factors carry the greatest predictive weight is therefore not simply an academic 443
question; it can help policymakers and public health officials determine where limited 444
resources may be most effectively directed to reduce preventable maternal morbidity and 445
mortality. However, because the relative importance of these factors varied across analyti- 446
cal scale, modeling approach, and state context, interventions may need to be tailored to 447
specific populations and contexts rather than organized around one universal "top factor." 448

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